What Reprices, and How Fast: The Composition Bias in Weighted-Average-Maturity Pass-Through

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Abstract

Analyses of how a rate shock reaches sovereign interest expenditure almost universally proxy the speed of pass-through with weighted average maturity, usually by assuming a constant repricing hazard whose expected repricing time equals the published figure. We show that this proxy is biased in a signable direction, quantify the bias, and identify its mechanism. For Portugal, a 100 basis-point shock reprices 10.90 percentage points more of the stock within one year than the proxy implies, worth about EUR 300 million of interest in the first year alone. The bias is predominantly *compositional* rather than behavioural: 14.2 percent of the stock is floating-rate or inflation-linked and reprices within a year regardless of maturity, and a maturity-calibrated hazard cannot represent that because those instruments reprice *without exiting*. We also test a behavioural channel, using Portuguese retail savings certificates, which are redeemable on demand and therefore a candidate for state-dependent pass-through. That test returns a well-measured null: the response to the competing-return spread is not distinguishable from zero, the predicted asymmetry has a bootstrap interval of [-0.020, +0.044], and the estimated kernel does not beat the standard benchmark out of sample. We report the null, and we report that the raw data contradicts the sign of the behavioural hypothesis: during the 2022–2023 tightening Portuguese households moved money *into* the certificates, not out.

1 Introduction

The speed at which a change in market yields reaches a sovereign's interest bill is governed by portfolio repricing. Practitioners and the official sector proxy that speed with weighted average maturity: a stock with an average maturity of m years is treated as repricing at a constant hazard $1/m$. The companion paper on Portugal's interest burden makes exactly this assumption, and it is the object this paper relaxes.

The proxy has a structural problem that has nothing to do with behaviour. A constant hazard treats the stock as homogeneous. Real sovereign portfolios are not: a material share is floating-rate or inflation-linked, and those instruments reprice on their own reset cycle *without leaving the stock*. A hazard calibrated to maturity cannot see them, because in a maturity model repricing and exit are the same event. They are not.

We quantify the resulting bias for Portugal, decompose it into a compositional component and a behavioural component, and test the behavioural component directly. Section 2 explains why Portugal permits that test at all. Section 3 sets out the kernel. Sections 4–5 describe what the published data does and does not identify. Sections 6–8 report the bias, the pass-through simulation, and an out-of-sample backtest.

2 Institutional setting

Portugal is an unusually good laboratory for a behavioural repricing test because a large, identifiable share of its debt is held in retail savings instruments — Certificados de Aforro and

Certificados do Tesouro — that households may redeem on demand subject to lock-ups and penalties. As of June 2026 these accounted for 15.4 percent of State direct debt. A holder facing a change in the competing return on bank deposits can act on it, so the repricing of that block is in principle a behavioural object rather than a contractual one.

The 2022–2023 tightening provides the episode. Certificados de Aforro grew from EUR 12.9 billion in June 2022 to EUR 32.6 billion by May 2023, a rise of 151 percent, peaking at a monthly inflow of EUR 3,549 million.

The direction matters and it is the reverse of the intuitive prediction. The behavioural argument for faster pass-through under tightening is that holders redeem and reinvest at the new rate. Portuguese households did the opposite: the certificate's remuneration formula tracked short rates, so tightening made it *out-compete* bank deposits and drew money in. Money arriving on the prevailing rate reprices the stock exactly as effectively as redemption and reissue, so the channel is real — but it operates on the subscription margin, not the redemption margin.

3 Framework

Let $K(h)$ be the share of the stock outstanding at the horizon origin that carries a repriced rate by horizon h. We build it from three components, combined rather than conflated.

Contractual. Debt that reprices by maturing and being reissued. Deterministic given a redemption schedule.

Reset. Floating-rate and inflation-linked debt, which reprices on its own cycle without exiting. *This is not a survival event*, and forcing it into a maturity model is the source of the compositional bias.

Behavioural. Retail money arriving on the prevailing rate. This component is a function of the shock, which is what makes the kernel state-dependent rather than a portfolio constant.

The benchmark against which we measure is the standard proxy,

$$K^{\mathrm{WAM}}(h) = 1 - e^{-h/m}, \tag{1}$$

with m the published weighted average maturity. Equation (1) is memoryless: it implies a long right tail, leaving a substantial share unrepriced after a decade.

4 Data

Instrument-class stocks and portfolio indicators come from IGCP historical series, monthly. Competing-return covariates come from the ECB Data Portal. The measurement layer for the aggregate debt and interest series is imported from the companion paper rather than recomputed.

Two limitations belong in the body, not an appendix.

The unit of observation is a class-month, not a holder. Retail certificate data is published only as an aggregate stock. An aggregate class-month panel is not an individual-level model and we do not describe it as one.

Only net stock is observed. Gross subscriptions and gross redemptions are not published. A redemption hazard is therefore not identified: a flat month is equally consistent with no activity and with large offsetting flows. We estimate the subscription margin, where positive net flow bounds the repriced amount from below, and we drop the redemption margin rather than fudging it. Months of net outflow are recorded as outflows, not as zero repricing.

No dated redemption schedule is published either, so the contractual track uses a linear retirement profile with the same mean. That is a stylised counterfactual shape, not the issuer's schedule, and the compositional result inherits the assumption.

5 Estimation

The specification was frozen before fitting and estimated once; no specification search was performed. The outcome is the repriced share of the opening stock; regressors are the signed parts of the lagged competing-return spread, an indicator for the June 2023 change in the terms offered on new subscriptions, and portfolio composition. Standard errors are Newey–West. With two instrument classes, cluster-robust inference is not valid and we do not report it.

The result is a null. The spread-widening coefficient is +0.0187 with a standard error of 0.0148 ($p = 0.20$). The asymmetry the design turns on — whether widening and narrowing load differently — has a bootstrap interval of [-0.020, +0.044] across 1,000 moving-block replicates and covers zero.

A falsification test supports reading this as imprecision rather than contamination: the fixed-rate share, a portfolio statistic no household observes, loads at -0.00019 ($p = 0.86$) and leaves the other coefficients essentially unchanged.

Three reasons for the imprecision, none repaired by a different specification. The sample contains one large widening episode, so its timing cannot be separated from anything else moving with it. The certificate's contractual remuneration rate is not published machine-readably, so a short-rate index stands in and attenuation of unknown size is expected. And the outcome is a one-sided bound, so net-outflow months contribute uninformative zeros.

6 The kernel and the bias

Table 1 reports the bias at a 100 basis-point shock. The sign is positive at every horizon: *the standard proxy understates how much of the stock reprices.*

Table 1: Bias in the weighted-average-maturity proxy, +100 bps

Horizon	Proxy share	Estimated share	Total bias (pp)	of which shape (pp)
1 year	0.1245	0.2335	10.90	7.44
3 years	0.3290	0.4168	8.78	-1.59
5 years	0.4857	0.5300	4.43	-5.85
10 years	0.7355	0.7641	2.86	-2.30

Source: author calculations. Shape and behaviour sum to the total.

The mechanism is compositional. At one year the shape component contributes 7.44 percentage points and requires no behavioural assumption whatsoever: 14.2 percent of the stock reprices within a year on its own reset cycle, and Equation (1) cannot represent it.

Two qualifications. First, the shape term is *not monotone* in the horizon: it turns negative by three and five years, because beyond the reset window the memoryless tail retires more than a profile that has already retired its fast component. "The geometric kernel is too slow" is true early and false later. Second, the behavioural component's band runs from zero at every horizon, because the estimate behind it is the null of Section 5. We report the central path only with that band; a point estimate without it would be precisely the failure this paper criticises.

In fiscal terms the one-year bias is 0.098 percent of GDP, about EUR 300 million.

7 Pass-through

Simulating forward under the estimated kernel corrects a further defect in the companion paper, which holds nominal GDP fixed across a ten-year horizon. At a 100 basis-point shock and a five-year horizon, the incremental burden is 0.475 percent of GDP under the frozen-GDP assumption and 0.391 percent under central nominal growth. Realistic growth removes about 18 percent of the measured effect.

We note one artefact so it is not mistaken for a finding. Simulated half-lives differ by shock direction, but that asymmetry is manufactured by clipping the behavioural response at zero, which prevents a rate fall producing a negative contribution. It is a modelling choice, and the estimated asymmetry was a null.

8 Backtest

We freeze the specification, estimate to a cut date, and predict forward using only information available at that date, feeding in realised issuance yields so that kernel error is isolated from yield-path error. The realised effective rate is imported from the companion paper.

Table 2: Out-of-sample mean absolute error, basis points on the effective rate

Cut date	Estimated kernel	Proxy benchmark
2014	52.44	55.69
2018	47.81	45.16
2021	14.24	14.66

The estimated kernel does not beat the benchmark. It wins narrowly at two cut dates, loses at one, and every margin is noise against the error levels. The 2021 cut places the entire tightening episode out of sample, which is where a behaviourally responsive kernel should have won most clearly.

This is consistent with the rest of the paper rather than in tension with it. The compositional bias is concentrated in the first year, and a backtest over four annual observations has little power to detect it against effective-rate volatility.

9 Discussion and limitations

The contribution is a measurement result: the standard pass-through proxy is biased, the bias is signable and quantified, and its mechanism is compositional. Anyone using weighted average maturity to time a rate shock through a budget should adjust for the share of the stock that reprices without exiting.

The behavioural thesis is not supported. We state that plainly. The channel is visible in the raw data — a 151 percent inflow and an abrupt stop at a dated policy change — but it is not identifiable at the precision published sources allow, principally because gross retail flows are not published and the contractual remuneration rate is not machine-readable.

Further limitations: the contractual track rests on a stylised retirement profile in the absence of a published schedule; the unit of observation is a class-month, not a holder; and the sample contains a single large tightening episode.

10 Conclusion

Weighted average maturity understates the speed at which a rate shock reaches a sovereign interest bill, by 10.90 percentage points of the stock at one year for Portugal, because a maturity-calibrated hazard cannot represent instruments that reprice without exiting. A behavioural channel exists and runs opposite to the intuitive direction, but the published data does not identify it.

Data and code availability

Every result in this paper is generated by the public repository at `https://github.com/DiogoRibeiro7/portugal-public-debt-interest`. The repricing panel is built with `pt-debt repricing build-panel` and the manuscript's numbers with `pt-debt repricing paper`, both against `config/repricing.yaml`. No number in the text above is typed by hand; the build fails if one is.

Cite the archived research object by its Zenodo concept DOI, DOI: 10.5281/zenodo.21722700, which covers all versions and resolves to the current one.

The companion interest-burden paper shares the measurement layer and is built from the same repository. A regression test asserts that the work reported here leaves that paper's outputs unchanged.

References

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